



Initial Setup (One-Time)

Install Git	<code>git-scm.com/downloads</code> → verify with <code>git --version</code>
Set Identity	<code>git config --global user.name "Your Name"</code> & <code>git config --global user.email "you@example.com"</code>
Verify	<code>git config --list</code>

Top 15 Git Commands

#	COMMAND	WHAT IT DOES	EXAMPLE & NOTES
1	<code>git init</code>	Create a new repository	<pre>mkdir my-project && cd my-project git init</pre> Initialized empty Git repository in ../.git/
2	<code>git status</code>	Check file status (tracked / modified / staged)	<pre>echo "Hello" > hello.txt git status</pre> 💡 Run this constantly — it's your best friend.
3	<code>git add</code>	Stage files for commit	<pre>git add hello.txt # single file git add . # all files git add *.txt # by extension</pre>
4	<code>git commit</code>	Save staged changes as a snapshot	<pre>git commit -m "Add hello.txt" git commit -am "Update files"</pre> 💡 Write meaningful messages!
5	<code>git log</code>	View commit history	<pre>git log --oneline git log --oneline --graph --all</pre>
6	<code>git diff</code>	See line-by-line changes	<pre>git diff # unstaged git diff --staged # staged</pre> 💡 Review before committing.
7	<code>git branch</code>	Create / list / delete branches	<pre>git branch feature-login # create git branch -d feature-login # delete</pre>
8	<code>git switch</code>	Switch branches (modern)	<pre>git switch feature-login # switch git switch -c feature-signup # create+switch</pre> Also works: <code>git checkout -b</code>
9	<code>git merge</code>	Combine branches	<pre>git checkout main git merge feature-login</pre> 💡 Switch to destination first, then merge.
10	<code>git remote</code>	Link to GitHub	<pre>git remote add origin https://github.com/USER/REPO.git git remote -v</pre>
11	<code>git push</code>	Upload commits to remote	<pre>git push -u origin main # first time git push # after that</pre>
12	<code>git pull</code>	Download & merge from remote	<pre>git pull git pull origin main</pre> 💡 Always pull before you push!
13	<code>git clone</code>	Copy a remote repository	<pre>git clone https://github.com/USER/REPO.git git clone -b branch https://github.com/USER/REPO.git</pre>
14	<code>git stash</code>	Temporarily save work	<pre>git stash # save git stash pop # restore git stash list # list all</pre>
15	<code>git reset</code>	Undo changes	<pre>git reset hello.txt # unstage git reset --soft HEAD~1 # undo commit, keep changes git reset --hard HEAD~1 # undo commit, DELETE changes</pre> ⚠️ --hard is dangerous! Use --soft to stay safe.

 Lab 1 — Push Your Code to GitHub

STEP	ACTION	COMMAND / DETAIL
1	Create GitHub repo	Go to github.com/new → name it <code>sahasra-project</code> → don't add README → Create
2	Initialize locally	<pre>git init git add . git commit -m "Initial commit: add project files"</pre>
3	Connect & push	<pre>git remote add origin https://github.com/YOUR-USERNAME/sahasra-project.git git branch -M main git push -u origin main</pre>
4	Verify	Open https://github.com/YOUR-USERNAME/sahasra-project — your files should be there!

 Lab 2 — Clone a Repository

STEP	ACTION	COMMAND
1	Clone	<pre>git clone https://github.com/YOUR-USERNAME/sahasra-project.git</pre>
2	Explore	<pre>cd sahasra-project ls git remote -v</pre>

 Lab 3 — Edit, Commit & Push

STEP	ACTION	COMMAND
1	Make a change	<pre>echo "New feature added" >> hello.txt</pre>
2	Review	<pre>git status git diff</pre>
3	Stage → Commit → Push	<pre>git add . git commit -m "Update hello.txt with new feature" git push</pre>

 Lab 4 — Work with Branches

STEP	ACTION	COMMAND
1	Create branch	<pre>git checkout -b feature-new-page</pre>
2	Make changes	<pre>echo "New page content" > newpage.txt git add newpage.txt git commit -m "Add new page"</pre>
3	Push branch	<pre>git push -u origin feature-new-page</pre>
4	Merge to main	<pre>git checkout main git merge feature-new-page git push</pre>

 Git Workflow

 Daily Routine

ORDER	COMMAND	PURPOSE
1	<code>git pull</code>	Get latest changes from remote
2	<i>...write your code...</i>	Do your work
3	<code>git add .</code>	Stage all changes
4	<code>git commit -m "msg"</code>	Save snapshot with message
5	<code>git push</code>	Upload to GitHub